

Multiple Choice

1. **Answer:** A

Options: A) Planck's constant; B) Boltzmann's constant; C) The Rydberg constant; D) The speed of light

Note: Correct option: A - Planck's constant

2. **Answer:** B

Options: A) 2; B) 250; C) 5,000; D) 10,000

Note: Correct option: B - 250

3. **Answer:** A

Options: A) 4; B) 5; C) 6; D) 20

Note: Correct option: A - 4

4. **Answer:** B

Options: A) 2; B) 250; C) 5,000; D) 10,000

Note: Correct option: B - 250

5. **Answer:** B

Options: A) 0.50 c; B) 0.60 c; C) 0.70 c; D) 0.80 c

Note: Correct option: B - 0.60 c

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '24.6 eV'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '2.5 ns'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Hall coefficient'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '3/2'.

Long Form Response

11. **Answer:** $0 < W_a < W_i$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** $1/3$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.